

Explainable AI-Based Heart Disease Risk Prediction System Using Ensemble Learning, LIME, and SHAP

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Abstract— Cardiovascular disease is a major global health concern, making early risk prediction and clinical decision support essential. The proposed Explainable Heart Disease Prediction System is a full-stack web-based framework that predicts heart disease risk using clinical parameters such as age, chest pain type, blood pressure, cholesterol, ECG results, maximum heart rate, exercise-induced angina, ST depression, major vessels, and thalassemia. The system uses preprocessing techniques such as feature scaling and categorical encoding with an ensemble stacking classifier built using Logistic Regression, Random Forest, and K-Nearest Neighbors. To improve transparency, LIME and SHAP are integrated to explain the major factors influencing each prediction. The system provides a risk score, risk category, visual feature contribution charts, clinical explanations, and personalized recommendations through a React frontend and FastAPI backend. This approach supports interpretable and user-friendly cardiac risk assessment.

Keywords— Heart Disease Prediction, Explainable Artificial Intelligence, Machine Learning, Ensemble Learning, Stacking Classifier, LIME, SHAP, Clinical Decision Support System.

I. INTRODUCTION

Cardiovascular disease is one of the major health challenges worldwide and remains a leading cause of mortality, making early prediction and preventive care highly important [7]. Heart disease risk is influenced by several clinical parameters such as age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, ECG results, maximum heart rate, exercise-induced angina, ST depression, number of major vessels, and thalassemia. Since these factors interact in complex ways, machine learning has become a useful approach for identifying hidden patterns in medical data and supporting clinical decision-making. Several studies have shown that machine learning models can effectively predict heart disease risk using patient health records [1], [3].

Traditional machine learning models such as Logistic Regression, K-Nearest Neighbors, and Random Forest have been widely used for heart disease prediction. However, the performance of individual models may vary based on the dataset, preprocessing methods, and feature relationships. To improve prediction reliability, ensemble learning techniques combine multiple models to produce a stronger final

prediction. Mohan *et al.* proposed a hybrid machine learning approach for effective heart disease prediction [1], while Noor *et al.* demonstrated the usefulness of stacking models with balancing techniques and dimensionality reduction [2]. Inspired by these works, the proposed system uses a stacking ensemble classifier consisting of Logistic Regression, Random Forest, and K-Nearest Neighbors.

Although machine learning can improve prediction accuracy, interpretability is essential in healthcare because doctors and users must understand why a prediction is made. Therefore, this project integrates Explainable Artificial Intelligence techniques such as LIME and SHAP to identify the major features influencing each prediction [5], [6]. Rezk *et al.* highlighted the importance of SHAP and LIME-based explanations in heart disease prediction systems [4]. The proposed system is developed as a full-stack web application using a React frontend and FastAPI backend, providing risk score, risk level, feature contribution charts, clinical explanations, and personalized recommendations. Thus, the system supports transparent and user-friendly cardiac risk assessment as a clinical decision-support prototype [8].

II. LITERATURE REVIEW

Machine learning-based heart disease prediction has been widely studied due to its ability to identify hidden relationships among clinical parameters and support early diagnosis. Mohan *et al.* [1] proposed an effective heart disease prediction approach using hybrid machine learning techniques. Their work demonstrated that combining different learning strategies can improve prediction performance compared with using a single classifier. Similarly, El-Sofany [3] investigated multiple machine learning classification techniques for predicting heart disease and highlighted the importance of selecting suitable algorithms and preprocessing methods for improving diagnostic accuracy. These studies show that machine learning can be effectively applied to structured medical datasets for cardiac risk prediction.

Ensemble learning has also received attention in heart disease prediction because it combines the strengths of multiple models to produce more reliable results. Noor *et al.* [2]

proposed a stacking-based heart disease prediction model with balancing techniques and dimensionality reduction. Their study showed that stacking models can improve prediction performance by integrating multiple classifiers and reducing the limitations of individual models. This supports the use of ensemble models such as Logistic Regression, Random Forest, and K-Nearest Neighbors in the proposed system, where the final prediction is generated through a stacking classifier.

Although several studies have focused on improving prediction accuracy, interpretability remains a major requirement in healthcare applications. A model that only predicts whether a patient has heart disease is not sufficient for practical clinical use, because doctors and users must understand the reasons behind the prediction. Ribeiro et al. [5] introduced LIME, an explanation technique that interprets individual predictions by identifying locally important features. Lundberg and Lee [6] proposed SHAP, which assigns contribution values to features based on Shapley theory. These techniques are especially useful in medical AI because they help explain how clinical factors such as cholesterol, blood pressure, chest pain type, and maximum heart rate influence the final prediction.

Recent research has combined heart disease prediction with explainable artificial intelligence. Rezk et al. [4] proposed an XAI-augmented ensemble model for heart disease prediction using SHAP and LIME. Their work emphasized that explainability improves trust and transparency in medical prediction systems. Fitriyani et al. [8] developed HDPM, a heart disease prediction model designed for clinical decision support, showing the importance of converting prediction models into usable healthcare support systems. These works highlight the need for systems that provide not only accurate predictions but also understandable explanations for healthcare professionals.

III. PROPOSED SYSTEM

A. Dataset

The proposed system uses a structured clinical dataset stored as heart.csv, containing patient attributes related to cardiovascular risk assessment. The dataset includes features such as age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, ECG result, maximum heart rate, exercise-induced angina, ST depression, number of major vessels, and thalassemia. The target variable target represents the disease outcome, where 0 indicates no heart disease and 1 indicates heart disease. Since the dataset contains both numerical and categorical attributes, preprocessing is applied before model training. The dataset is divided into training and testing subsets to train the model and evaluate its performance on unseen patient records.

B. Data Preprocessing

To prepare the clinical dataset for machine learning-based heart disease prediction, preprocessing is applied to convert

raw patient attributes into a suitable format [1], [3]. The following steps are used:

- **Feature and Target Separation:** The dataset is separated into input features X and target variable y , where 0 indicates no heart disease and 1 indicates heart disease.
- **Numerical Feature Scaling:** Numerical features such as age, blood pressure, cholesterol, maximum heart rate, and ST depression are scaled using StandardScaler.
- **Categorical Feature Encoding:** Categorical features such as sex, chest pain type, ECG result, exercise-induced angina, slope, major vessels, and thalassemia are encoded using OneHotEncoder.
- **Column-wise Transformation:** A ColumnTransformer applies scaling and encoding separately to the appropriate feature groups.
- **Pipeline Integration and Splitting:** The preprocessing steps are integrated with the classifier using a Pipeline, and the dataset is split into training and testing subsets to evaluate performance on unseen records [8].

C. Model Architecture

The proposed system follows a full-stack machine learning architecture consisting of three main modules: the frontend input module, the backend prediction module, and the explainable AI output module (Fig. 1).

1. **Frontend Input Module:** The user enters patient clinical details through a React-based web interface. The input form includes doctor-friendly fields such as age, sex, chest pain type, resting blood pressure, cholesterol, ECG result, maximum heart rate, exercise-induced angina, ST depression, major vessels, and thalassemia. Categorical inputs are represented using dropdown options instead of raw numerical codes to improve usability.
2. **Machine Learning Prediction Module:** The backend is developed using FastAPI and receives patient data through a REST API. The input data is processed using a preprocessing pipeline that applies StandardScaler for numerical features and OneHotEncoder for categorical features. The prediction model is a stacking ensemble classifier that combines Logistic Regression, Random Forest, and K-Nearest Neighbors as base learners, with Logistic Regression used as the final estimator. Ensemble-based learning is used to improve prediction reliability by combining the strengths of multiple classifiers [1], [2].
3. **Explainable AI and Recommendation Module:** After prediction, the system calculates a risk score and classifies the patient into Low, Medium, or High risk. To improve transparency, LIME and SHAP are used to identify the important clinical features contributing to the prediction [5], [6]. The results are displayed using visual charts, clinical explanations, and personalized recommendations. This makes the

system more interpretable and suitable as a clinical decision-support prototype [4], [8].

D. Libraries and Frameworks

Implementation of the proposed Heart Disease Prediction System uses the following technologies:

- **Python:** Used as the core programming language for backend development, machine learning, data processing, and explainable AI integration.
- **Pandas and NumPy:** Used for dataset loading, data manipulation, numerical computation, and converting patient inputs into machine learning-compatible format.
- **Scikit-learn:** Used for preprocessing, pipeline construction, model training, and development of the stacking ensemble classifier using Logistic Regression, Random Forest, and K-Nearest Neighbors [1], [2].
- **LIME and SHAP:** Used to explain individual predictions by identifying the clinical features that contribute most to the predicted heart disease risk [5], [6].
- **FastAPI:** Used to build the backend REST API that receives patient data, performs prediction, and returns the risk score, explanations, and recommendations.
- **React:** Used to develop the frontend web interface where users enter patient details and view prediction results interactively.
- **Axios:** Used to enable communication between the React frontend and FastAPI backend through REST API calls.
- **Recharts and Bootstrap:** Recharts is used to visualize LIME and SHAP feature contributions, while Bootstrap is used to create a responsive and user-friendly interface.

E. System and Implementation

The system architecture of the proposed Heart Disease Prediction System is modular, ensuring smooth data flow between the frontend interface, backend prediction engine, machine learning pipeline, explainability module, and result visualization components. The process begins with the user input interface, where patient clinical details such as age, sex, chest pain type, resting blood pressure, cholesterol, fasting blood sugar, ECG result, maximum heart rate, exercise-induced angina, ST depression, major vessels, and thalassemia are entered through a React-based web application. To improve usability, categorical medical values are represented using meaningful dropdown options instead of raw numerical codes.

The entered patient data is sent to the backend through REST API communication using Axios. The backend is implemented using FastAPI, which receives the input data, validates it, and passes it into the machine learning pipeline. The preprocessing module applies feature scaling to numerical attributes using

StandardScaler and converts categorical attributes into machine-readable form using OneHotEncoder. A ColumnTransformer is used to apply these transformations separately, and the preprocessing stage is integrated with the classifier through a Scikit-learn Pipeline. Such preprocessing and classification workflows are commonly used in machine learning-based heart disease prediction systems to improve data consistency and prediction reliability [1], [3], [8].

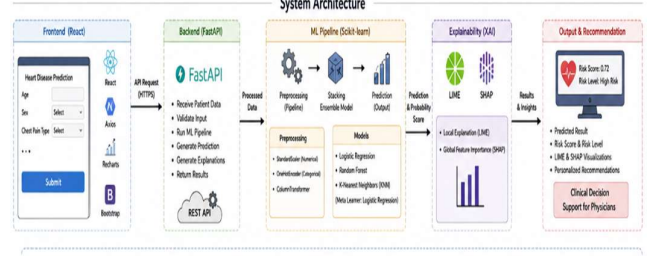


Fig. 1. System architecture of the proposed Explainable Heart Disease Prediction System.

The prediction module uses a stacking ensemble classifier consisting of Logistic Regression, Random Forest, and K-Nearest Neighbors as base learners, with Logistic Regression used as the final estimator. Ensemble and hybrid learning approaches have been shown to improve heart disease prediction performance by combining the strengths of multiple classifiers [1], [2]. The model predicts the probability of heart disease and converts it into a risk score and risk category such as Low, Medium, or High. To improve model transparency, the explainability module applies LIME and SHAP to identify the most influential clinical features contributing to each prediction [5], [6]. The final results, including prediction output, risk score, risk level, top contributing factors, visual explanation charts, clinical explanations, and personalized recommendations, are returned to the React frontend. This makes the proposed framework suitable as an explainable clinical decision-support prototype [4], [8].

F. Algorithmic Workflow

The proposed Heart Disease Prediction System follows a structured workflow that integrates clinical data input, preprocessing, machine learning prediction, explainable AI analysis, and web-based result visualization. The process begins when the user enters patient health details through the React frontend, including age, chest pain type, blood pressure, cholesterol, ECG result, maximum heart rate, exercise-induced angina, ST depression, major vessels, and thalassemia. These inputs are sent to the FastAPI backend through a REST API, where the data is converted into a structured format suitable for prediction. The preprocessing pipeline then applies feature scaling to numerical attributes and one-hot encoding to categorical attributes using Scikit-learn transformers [1], [3].

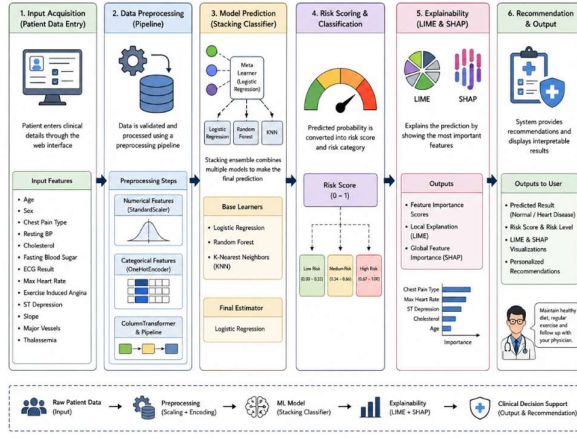


Fig. 2. Algorithmic workflow of the proposed Heart Disease Prediction System.

After preprocessing, the transformed data is passed to a stacking ensemble classifier consisting of Logistic Regression, Random Forest, and K-Nearest Neighbors as base learners, with Logistic Regression used as the final estimator [2]. The model predicts the probability of heart disease and converts it into a risk score and risk category such as Low, Medium, or High. To improve transparency, LIME and SHAP are applied to identify the major clinical factors influencing the prediction [5], [6]. The final output, including prediction result, risk score, top contributing features, visual explanation charts, clinical interpretation, and personalized recommendations, is returned to the React frontend. This workflow supports transparent and user-friendly cardiac risk assessment by combining prediction accuracy with explainability and clinical usability [4], [8].

IV. RESULTS AND DISCUSSION

The proposed Heart Disease Prediction System was implemented as a full-stack web application using React for the frontend and FastAPI for the backend. The machine learning pipeline was developed using Python, Scikit-learn, LIME, and SHAP. The patient input interface was designed with clinically understandable labels and dropdown options for categorical attributes such as sex, chest pain type, ECG result, exercise-induced angina, slope, and thalassemia, as shown in Fig. 3. This improves usability by avoiding raw numerical codes and allowing users to enter patient details in a more meaningful format.

After submitting the patient details, the backend preprocesses the input and passes it to the stacking ensemble classifier. The system then generates the predicted heart disease outcome, risk score, and risk level. As shown in Fig. 4, the result is presented as a triage-style assessment that includes the prediction, risk category, primary risk drivers, clinical logic breakdown, and personalized next steps. Instead of providing only a binary result, the system converts the predicted probability into a percentage-based risk score and classifies it as Low, Medium, or High risk. Such risk-based output is

useful in clinical decision-support systems because it provides a clearer interpretation of patient risk severity [8].

Fig. 3. Patient clinical input interface.

Fig. 4. Prediction output with risk score, explanation, and recommendations.

The explainability module improves transparency by using LIME and SHAP to identify the clinical features that influence the prediction [5], [6]. The LIME visualization shown in Fig. 5(a) explains the local feature contributions for an individual patient, while the SHAP visualization shown in Fig. 5(b) presents feature impact values that indicate whether a factor increases or decreases the predicted risk. In the tested output, features such as major vessels, thalassemia, chest pain type, ST depression, and maximum heart rate were identified as important contributors. Previous studies have also emphasized that explainable AI methods such as LIME and SHAP improve trust and interpretability in heart disease prediction systems [4].



Fig. 5. LIME and SHAP-based explainability visualizations.

The recommendation section provides personalized guidance based on major risk factors identified from the patient input and explanation results. For example, elevated cholesterol, high resting blood pressure, and increased oldpeak values generate suggestions related to diet, exercise, monitoring, and medical follow-up. These recommendations are not intended to replace professional diagnosis but serve as supportive guidance for risk awareness. Overall, the results show that the proposed system successfully combines heart disease prediction, risk scoring, explainable AI visualization, and recommendation generation within a single user-friendly clinical decision-support prototype.

V. CONCLUSION AND FUTURE SCOPE

The proposed Explainable Heart Disease Prediction System successfully demonstrates the integration of machine learning, explainable artificial intelligence, and full-stack web development for cardiac risk assessment. The system uses a stacking ensemble classifier built using Logistic Regression, Random Forest, and K-Nearest Neighbors to predict heart disease risk from clinical patient attributes. The preprocessing pipeline applies numerical scaling and categorical encoding to prepare the dataset for accurate classification. The final prediction is presented as a risk score and risk level, making the result more understandable than a simple binary output. Ensemble-based learning has been shown to improve heart disease prediction performance by combining the strengths of multiple classifiers [1], [2].

A major strength of the proposed system is its use of explainable AI techniques. LIME and SHAP are integrated to identify the important clinical features influencing each prediction, improving transparency and interpretability [5], [6]. The web interface displays the prediction result, risk level, primary risk drivers, clinical explanation, feature contribution charts, and personalized recommendations. This makes the

system suitable as a clinical decision-support prototype, where users can understand not only the predicted outcome but also the factors that contributed to the result. Previous studies have also highlighted the importance of explainability and clinical decision support in healthcare-based prediction systems [4], [8].

In future work, the system can be improved by training and validating the model on larger and more diverse medical datasets to increase generalization. Additional features such as patient history management, doctor login, database integration, downloadable medical reports, and cloud deployment can be added to improve practical usability. The recommendation module can also be enhanced using expert-defined medical guidelines or counterfactual explanation techniques, allowing the system to suggest possible changes that may reduce predicted risk. Overall, the proposed work represents a practical step toward transparent, user-friendly, and explainable AI-based heart disease risk prediction.

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